There is no sound and complete derivation system for second-order logic

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Editorial scope. Statement and complete author proof are included verbatim as the span lines 731-756 of the pinned original source; no text is rewritten or paraphrased.

Allowed standard prerequisites (editorial).

- second-order arithmetic PA^2 and its semantic entailment
- Tarski's theorem on the undefinability of truth
- representability of computable functions in Q

Theorem 0.1. *There is no sound and complete !!derivation system for second-order logic.*

Proof. Let $!A$ be !!asentence in the language of arithmetic. $N \models !A$ iff $\mathrm{Th}PA^2 \models !A$. Let $!P$ be the conjunction of the nine axioms of $\mathrm{Th}PA^2$. $\mathrm{Th}PA^2 \models !A$ iff $\models !P \rightarrow !A$, i.e., $M \models !P \rightarrow !A$. Now consider the !!sentence $\forall[z][\forall[u][\forall[u'][\forall[u'']][\forall[L][(!P' \rightarrow !A')]]]]$ resulting by replacing 0 by z, $'$ by the one-place function variable u, $+$ and $\times$ by the two-place function-variables u' and u'', respectively, and $<$ by the two-place relation variable L and universally quantifying. It is a valid sentence of pure second-order logic iff the original sentence was valid iff $\mathrm{Th}PA^2 \models !A$ iff $N \models !A$. Thus if there were a sound and complete proof system for second-order logic, we could use it to define a computable enumeration $f\colon \mathbb{N} \to \mathrm{Sent}[L_A]$ of the !!sentences true in $\mathfrak{A}N$. This function would be representable in ThQ by some first-order formula $!B_f(x, y)$. Then the !!formula $\exists[x][!B_f(x, y)]$ would define the set of true first-order !!sentences of $\mathfrak{A}N$, contradicting Tarski's Theorem. □

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